WELA

Windows Event Log Analyzer

Technical reference & demo guide

How every verdict is made — power cycles, the drive's part in them, whether that drive is a Samsung drive, and which app, update or user triggered them — with the reasons behind each design decision, the demo script, the limitations, the roadmap and where AI fits in.

QUESTION 1
How did each power cycle end?

QUESTION 2
Did the drive cause it?

QUESTION 3
Which drive — is it the Samsung drive?

QUESTION 4
What triggered it — an app, an update, the user?

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A Appendix: event reference
Every event ID WELA reads

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Run, test, collect, configure

How to read this. Each chapter opens with a one-line takeaway (blue bar). Tables hold the exact rules — every number in them comes from `wela/catalog.py` and `wela/engine.py` as committed on this branch. Chapter 17 is the demo script for the meeting; chapter 16 lists the caveats to state up front.

1 At a glance

> WELA turns a claim's Windows event logs into a verified, explainable story of every power cycle — in seconds, on the engineer's own PC.

The 30-second pitch. Customers return SSDs after crashes, freezes and sudden power-offs. The PC's own Windows event logs hold the evidence, but they are thousands of events spread over several logs. WELA reads them and, in seconds, rebuilds every power cycle; shows whether the drive was misbehaving right before it; whether that drive was the Samsung drive; and whether an app, an update or the user triggered it. Every verdict comes with the events behind it and a confidence score, and anything uncertain is flagged for review.

The four questions WELA answers

Question	What WELA reports	Built from
1 · How did it end?	Normal shutdown · restart · Fast Startup · hard shutdown (crash, power-button long-press, power lost while asleep, hang during shutdown, power loss) · sleep / hibernate / Modern Standby · undetermined	Kernel-General 12/13, EventLog 6005/6006/6008, Kernel-Power 41/42/107/109, User32 1074, WER 1001, Power-Troubleshooter 1
2 · Did the drive cause it?	Likely · Possibly · No drive evidence · Can't tell — plus the effect on the drive after boot	Storport (every level), disk, stornvme/storahci, NTFS, volmgr, WHEA, crash (bugcheck) codes
3 · Which drive?	Samsung named or mapped · Samsung? probably · other drive · not identified	Storport VendorId/ProductId/SerialNumber, Partition 1006 disk inventory per boot, PCI vendor IDs
4 · What triggered it?	Windows Update · installer · OEM/vendor tool · IT tool · script · user · system-process crash · update still installing · apps hung · app crash — as Confirmed Likely Possibly	User32 1074 + its reason code, Windows Update, MsiInstaller, Application Error/Hang, Windows Error Reporting, Service Control Manager, Display

Key numbers

0 installs
Python standard library only; `.evtx` read with Windows' own `wevtutil`

100 % local
UI served on 127.0.0.1 only; no internet; no data leaves the PC

~4.7 s
to analyse ~200,000 events (6,400 power cycles) — analysis step, measured on a laptop

51 tests
rule tests, a full demo scenario with known answers, and a random-input robustness test

Every verdict
lists the events found, the expected events not logged, its reasons and a confidence score

4 export formats
HTML report, Excel workbook (7 sheets), CSV and JSON

Design principles

Principle	What it means in practice
Explainable	Nothing is a black box: each verdict shows the evidence, the missing evidence, plain-language reasons and a confidence score. Rules live in one readable file.
Conservative	WELA says "undetermined", "can't tell" or "not identified" instead of guessing. A quiet log is never presented as proof the drive is fine.
Private	Runs on the engineer's PC; the web server binds to 127.0.0.1; no cloud, no telemetry.

Principle	What it means in practice
Zero-install	Standard-library Python 3.9+; no packages needed on Windows. Works offline.
Tunable	Every event ID, weight, threshold and vendor rule is in `catalog.py` and can be overridden with a JSON file — no engine changes.
Robust	Unreadable files, missing logs, record gaps, clock changes and malformed events are reported, not fatal.

2 Why WELA exists

> Claim analysis needs the same evidence-based answer every time — reading Event Viewer by hand is slow, inconsistent and easy to get wrong.

The problem

When an SSD comes back under claim, the key questions are about **power cycles**: did the PC shut down cleanly, crash, or lose power? Was the drive already failing before (the drive *caused* it) or did the problem start after (the event *affected* the drive)? Was it even our drive — or another disk, a USB stick, the platform? Or was the restart simply Windows Update?

All of that is in the Windows event logs, but:

- **Volume** — a claim's logs hold tens of thousands of events across System, Application, Storport and Partition logs.
- **Scattered evidence** — one hard shutdown leaves traces in three logs (e.g. Storport warnings, disk retries, then Kernel-Power 41 and EventLog 6008 after the next boot).
- **Missing evidence matters** — the *absence* of a stop event (no 1074/13/6006) is itself evidence, and so are lost or cleared records. Humans rarely check this systematically.
- **Inconsistency** — two engineers reading the same logs can reach different conclusions; manual review of one claim can take hours.
- **Attribution** — a "Reset to device" event does not say which drive; mapping it to the Samsung drive needs the disk inventory and care.

The approach

Automate the reading. Merge every log into one UTC timeline, find every boot, and decide how the previous session ended — using redundant evidence and timing, not one event.

Correlate around each power cycle. Look at the drive in the minute before power-off and after boot, and at every app and update event from the power cycle before to the one after.

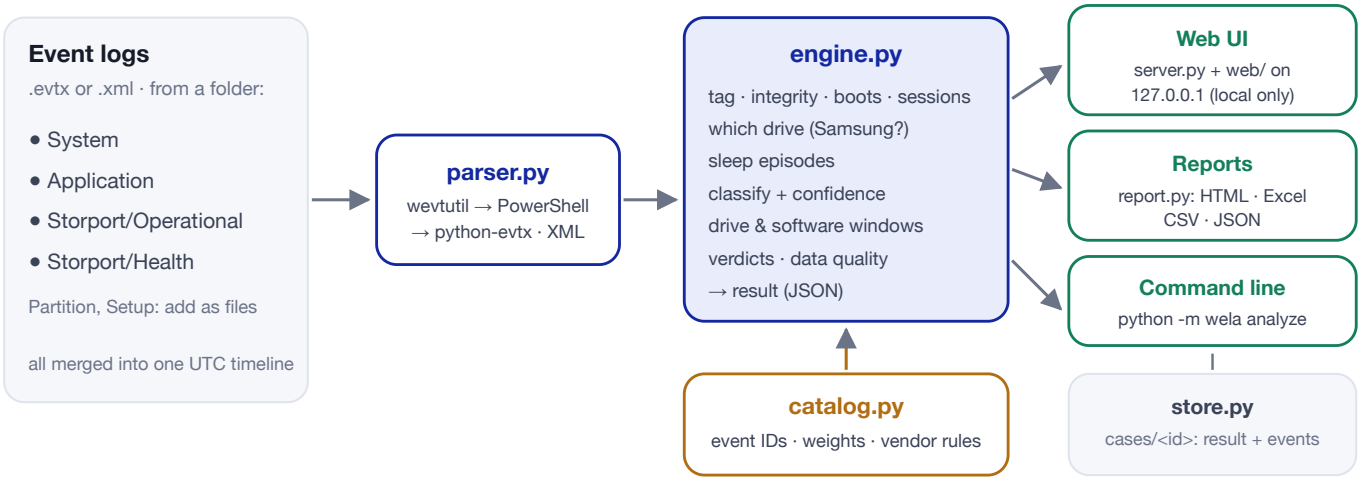
Attribute carefully. Tie each storage event to a physical drive and say how that link was made — named, mapped, probable or unknown.

Show the work. Confidence scores, missing-evidence lists and review flags make uncertainty visible instead of hiding it.

It complements, not replaces, the drive's own SMART and error logs: a quiet Windows log never clears the drive (chapter 8).

3 Architecture

> A small, single-language codebase: a parser, a rule catalog, an analysis engine, and three ways to see the result (web UI, reports, command line).



Components

File	Responsibility	Lines
`wela/parser.py`	Reads `.evtx` (wevtutil, then PowerShell, then the optional python-evtx package on non-Windows) and XML exports into one normalised event record: UTC time, log, provider, event ID, record ID, level, data fields. From a folder it picks only System, Application and Storport (Operational, Health).	236
`wela/catalog.py`	The knowledge base: every power signal, storage and software event ID, evidence weight, threshold, crash code, vendor rule and shutdown-initiator pattern.	282
`wela/engine.py`	The analysis pipeline (chapter 5): from timeline to verdicts, confidence, drive identity, software trigger and the app / update lists.	1,470
`wela/report.py`	Self-contained HTML report, Excel workbook written directly as Office Open XML (no library), CSV and JSON.	406
`wela/server.py`	Local HTTP server and JSON API; native file / folder picker.	232
`wela/store.py`	Saves each case under `./cases/<id>` (result JSON + compressed events for the "context" viewer).	84
`wela/demo.py`	Synthetic demo logs covering every tricky scenario, with the expected answers the tests check.	312
`web/ (app.js, app.css, index.html)`	The UI — vanilla JavaScript and CSS, no frameworks or CDN, works offline; light and dark themes.	1,173
`tests/test_engine.py`	51 automated tests (chapter 14).	535

Technology choices — and why

Choice	Reason
Python standard library only	Nothing to install on locked-down Windows PCs; simpler IT approval; one language for parser, engine, server and reports.
Windows' own `wevtutil`	Microsoft's reader for `.evtx` — exact field names and values, no third-party parser. PowerShell and python-evtx are fallbacks.
Local web UI on 127.0.0.1	A modern browser interface without deploying a server; claim data never leaves the PC.

Choice	Reason
Rules in a catalog, not in code	Reviewers can audit and tune event IDs, weights and thresholds; a JSON override changes behaviour without a code change.
Deterministic rules, not a model	Claim decisions must be reproducible and auditable: the same logs always give the same verdict, with the evidence shown. (AI is proposed on top, chapter 19.)
Excel written directly	Office Open XML is a zip of XML files; writing it directly avoids an external dependency while still giving frozen headers, filters and highlighted rows.

Local API (used by the UI)

`GET /api/cases` · `GET /api/cases/<id>` · `GET /api/cases/<id>/events?around=<idx>&n=20` · `GET /api/cases/<id>/events?tag=software&from=…&to=…` (a whole app / update list) · `GET /api/cases/<id>/export.(html|xlsx|csv|json)?tz=330` · `POST /api/analyze` · `POST /api/demo` · `POST /api/pick` · `DELETE /api/cases/<id>` · `GET /api/config`

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4 Inputs: the logs

> System is required; Application and Storport make the drive and software answers strong. From a folder WELA reads only these — the Partition log, which ties disk events to a drive, is added as a file.

Log	Collect with (admin command prompt)	What WELA uses it for	Needed?
System	`wevtutil epl System System.evtx`	Boots and shutdowns (12/13, 6005/6006/6008, 41, 109, 1074), sleep, disk / stornvme / NTFS / volmgr / WHEA events, Windows Update, service failures, display driver timeouts	Required
Application	`wevtutil epl Application Application.evtx`	App crashes and hangs, Windows Error Reporting (incl. blue-screen reports), MSI installer events	Strongly recommended
Storport	`wevtutil epl Microsoft-Windows-Storage-Storport/Operational Storport.evtx`	Every Storport event at every level (timeouts, resets, latency reports); these events usually **name the drive** (vendor, product, serial)	Strongly recommended
Storport Health	`wevtutil epl Microsoft-Windows-Storage-Storport/Health Storport-Health.evtx`	Storport's health log, read the same way as Operational: every event at every level, tied to the drive it names	Recommended
Partition	`wevtutil epl Microsoft-Windows-Partition/Diagnostic Partition.evtx`	Disk inventory at every boot (model, firmware, serial, disk number, bus type) — maps `\Device\HarddiskN` to a drive	Optional add as a file
Setup	`wevtutil epl Setup Setup.evtx`	Windows servicing "a restart is needed" events	Optional add as a file

- Give WELA a folder (searched recursively) or individual files; `.evtx` and `.xml` both work, and files can be mixed.
- **From a folder only System, Application and Storport (Operational, Health) are read**, so a copy of `C:\Windows\System32\winevt\Logs` with hundreds of logs works as it is. Files are recognised by name — as winevt\Logs names them (`Microsoft-Windows-Storage-Storport%4Health.evtx`) or as exported (`Storport.evtx`, `PC01_System.evtx`); the rest are skipped and counted. A file added by itself is always read.

- All events are merged into one timeline in UTC; the UI and reports convert to the chosen time zone (UTC, IST, local, US, CET, China, Korea).
- A file that cannot be read is listed under Data quality with the reason; the rest of the analysis continues.
- Missing logs are reported ("Missing: Storport log — related checks are limited") and the affected verdicts say so.

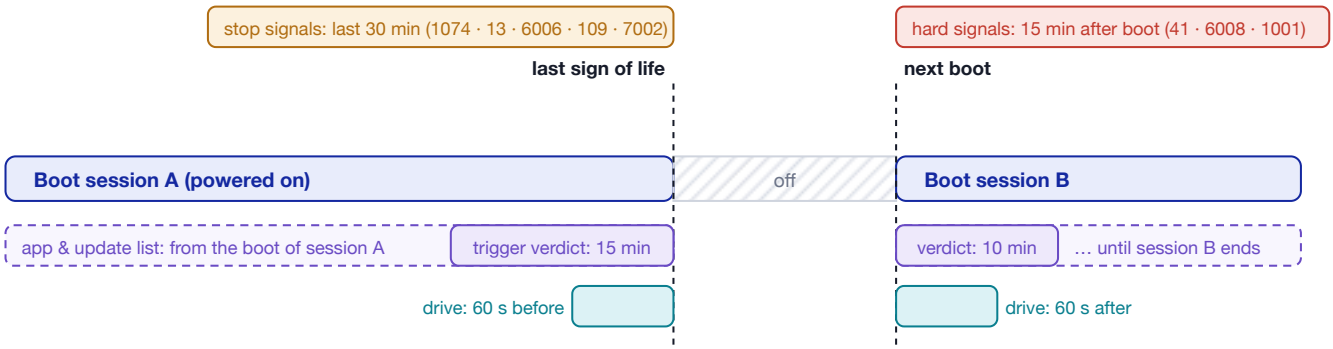
Collect early. Operational and diagnostic logs such as Storport and Partition can be small and roll over; ask for the logs as soon as possible after the incident, and before any reinstall.

5 The analysis pipeline

> Twelve deterministic steps; each one only adds information, so every verdict can be traced back to the events it used.

#	Step	What happens	Why
1	**Parse & merge**	Every event becomes one record (UTC time, log, provider, ID, record ID, level, data). All logs are sorted into one timeline; from a folder only the four logs of chapter 4 are read.	Evidence for one power cycle sits in several logs.
2	**Tag**	Each event gets up to three meanings from the catalog: a power signal (e.g. "OS started"), a storage meaning (e.g. "Reset to device", state *reset*), a software meaning (e.g. "Application hung").	Rules work on meanings, not raw IDs; the catalog is the single place to change them.
3	**Integrity**	Coverage of each log, gaps in record IDs, cleared logs, time going backwards.	Missing records can hide the stop events — verdicts near gaps must be weaker.
4	**Boots**	Kernel-General 12, EventLog 6005/6009 and Kernel-Boot 27 within 3 minutes count as one boot. BootType=2 is a hibernate resume, not a boot.	Redundant markers: a boot is found even if one marker is missing.
5	**Sessions**	Everything between two boots is one boot session.	Each boot closes the previous session — that ending is what we classify.
6	**Which drive**	Each storage event is tied to a physical drive (chapter 9).	"Reset to device" is only meaningful if we know whose device.
7	**Sleep**	Sleep, hibernate and Modern Standby episodes rebuilt from 42 / 107 / Power-Troubleshooter 1 / 506 / 507 / 27.	Any one of them is enough to rebuild an episode.
8	**Classify**	For every boot: how did the previous session end, with a confidence score (chapters 6–7).	The core answer to question 1.
9	**Correlate**	Drive window: 60 s before the last logged event and 60 s after boot. App / update lists: from the power cycle before to this one, and from this one to the next; the trigger verdict reads the 15 min before and the 10 min after.	Cause shows before power-off; effect after boot; a slow cause can start long before.
10	**Assess**	Did the drive cause it? Effect on the drive. Which drive had the problems. What triggered it (software).	Questions 2, 3 and 4.
11	**Data quality**	Big clock changes, long unexplained quiet gaps inside a session.	Context for reviewers (chapter 11).
12	**Result**	Counts, power-cycle list, storage and software event lists, disk inventory, integrity report — saved as one JSON per case.	Feeds UI, reports, CLI and, later, AI.

Anatomy of one power cycle — the windows WELA looks at



Not to scale. For sleeps the windows sit around sleep entry and wake; lists can also start or end at a sleep. Lengths are settings in catalog.py.

Why "last sign of life"? After a hard shutdown there is no shutdown event to anchor on, so WELA uses the last event any log recorded before the next boot. "−17 s" next to a drive event means it was logged 17 seconds before that point.

6 How a power cycle is classified

> For every boot, WELA asks: did the previous session stop cleanly, did Windows report an unclean stop, or is the evidence missing?

Evidence	Verdict
1074 / 109 / 13 / 6006 in the last 30 min of the session	Clean stop → **restart** (1074 says restart, or the next boot follows within 5 min) or **shutdown**
Kernel-Boot 27 with BootType = 1 on the next boot	The shutdown was **Fast Startup** (kernel hibernated)
Kernel-Power 41 and/or EventLog 6008 after the boot (+ 1001)	Hard shutdown — subtype from the 41 fields below
Stop started (1074) but then 41	Shutdown **hung** and did not complete
No stop events, no 41 / 6008	**Inferred** hard shutdown — low confidence, flagged for review
Record gap at the boundary, or the log starts at this boot	Undetermined — not guessed

Hard-shutdown subtype (Kernel-Power 41 fields)

Field	Subtype
`BugcheckCode` ≠ 0 (or WER 1001)	Crash (bugcheck 0x…)
`PowerButtonTimestamp` ≠ 0	Power-button long-press
`SleepInProgress` ≠ 0, or an open sleep	Power lost during sleep / hibernate
Stop events but 41	Shutdown started but did not complete (hang)
None of the above	Power loss / hang / forced reset

Restart or shutdown?

1. 1074's shutdown type (parameter 5) — matched in many languages: English, German, French, Spanish, Italian, Dutch, Japanese, Chinese, Korean, Russian.
2. Next boot is a Fast Startup boot → shutdown.
3. Otherwise: stop → boot gap under 5 min → restart (flagged as "inferred from gap").
4. 1074 says restart but the boot came > 30 min later → warning.

Sleep, hibernate, Modern Standby

Kernel-Power 42 (enter, TargetState 5 = hibernate) → 107 (resume) → Power-Troubleshooter 1 (carries sleep *and* wake time, plus wake source). Any one is enough to rebuild the episode. Modern Standby uses 506/507; hibernate resume uses Kernel-Boot 27 BootType = 2. A sleep with no resume followed by a fresh boot means power was lost while asleep.

Conflicts are flagged, not hidden. A clean stop (13 + 6006) followed by "unclean shutdown" on the next boot loses confidence and raises a "check manually" warning; so do restarts that took far longer than a restart should.

7 Confidence scoring

> Confidence = the sum of the weights of the evidence found (capped at 100 %), minus penalties for conflicts and log gaps. Below 60 %, or any warning, means "Needs review".

Clean-stop evidence	Weight
Shutdown requested (User32 1074)	0.35
OS stopped (Kernel-General 13)	0.30
Event log stopped (6006)	0.30
Kernel shutdown initiated (Kernel-Power 109)	0.20
User logoff (Winlogon 7002)	0.05
1074 states the type (restart / power off)	+0.10

Hard-shutdown evidence & penalties	Weight
Unclean shutdown reported (Kernel-Power 41)	0.50
Previous shutdown unexpected (6008)	0.40
Bugcheck report (1001)	0.20
Session ended with no stop signal at all	0.30
Conflict: clean stop logged, yet unclean on boot	−0.25
Record gap at this boundary	−0.30

Worked examples (from the demo case)

ID	Situation	Calculation	Result
T2	Normal shutdown	1074 0.35 + 13 0.30 + 6006 0.30 + 109 0.20 + 7002 0.05 = 1.20, capped at 1.0; + 0.10 for the stated type → capped again	**100 %**
T13	Restart, no 1074	13 0.30 + 6006 0.30 + 109 0.20 + 7002 0.05 = 0.85; restart inferred from a 1-min gap (info flag)	**85 %**
T5	Hard shutdown, power loss	41 0.50 + 6008 0.40 + no stop signal 0.30 = 1.20 → capped	**100 %**
T10	Session just ends, nothing logged	no stop signal 0.30 only — "Inferred" + warning	**30 %** · review
T11	37 System records missing at the boundary	Undetermined, fixed 0.20 — the stop events may be in the gap	**20 %** · review

Sleep episodes score by how many of the three sleep signals (42 / 107 / Power-Troubleshooter 1, or 27 for hibernate) were found: none 30 %, one 60 %, two 85 %, all 100 %; Modern Standby 90 % with both 506 and 507, else 50 %; a sleep that never resumed is capped at 50 %.

Three separate confidence measures. The percentage on a power cycle is about *how it ended*. "What triggered it?" has its own percentage (chapter 10). Drive verdicts use labelled levels (Likely / Possibly / No drive evidence / Can't tell) and drive attribution uses sure vs. probable (chapter 9) — each answers a different question, so they are kept apart.

8 Did the drive cause it?

> Drive trouble in the minute before power-off points at the drive as a cause; trouble in the minute after boot shows the effect on the drive.

What is checked

Every storage event — every Storport event at every level, plus disk, stornvme / storahci / Intel RST resets, NTFS, volmgr and WHEA — in the 60 s before the last logged event and the 60 s after boot (for sleeps: before sleep entry and after wake). Each is listed with its offset and key fields; SCSI opcodes are decoded (e.g. `Opcode=42 (WRITE(10))`). Informational events are listed but do not change the verdict.

Drive state (worst first)	Example events
missing	disk 157 — disk surprise-removed
reset	stornvme / storahci 129 — reset to device
error	disk 7, 11, 51, 52, 154; NTFS 50/55/98/140; WHEA 18
slow	disk 153 — I/O retried
warning	Storport warnings; WHEA 17/19/47 corrected errors

Crash code	Meaning	Points at storage
0x7A · 0x77	Kernel data / stack in-page error	strongly
0x7B · 0xED	Inaccessible boot device / unmountable volume	strongly
0xF4	Critical object termination	strongly
0x24 · 0x9F · 0xEF	NTFS · driver power state · critical process died	sometimes
0x133 · 0x154	DPC watchdog · unexpected store exception	sometimes

The verdict (hard shutdowns and undetermined power cycles)

Verdict	Rule
Likely	A drive reset, error or surprise removal in the 60 s before power-off, or a crash code that strongly points at storage.
Possibly	Slow I/O or warnings before power-off; platform (WHEA) errors only — they may come from another device; or a crash code that is sometimes caused by storage.
No drive evidence	A log covers the window and the drive was quiet.
Can't tell	No log covers the window.
No (orderly)	Clean shutdowns and restarts — the drive did not cause them; drive errors during the shutdown are still listed ("check for failed writes or flushes").

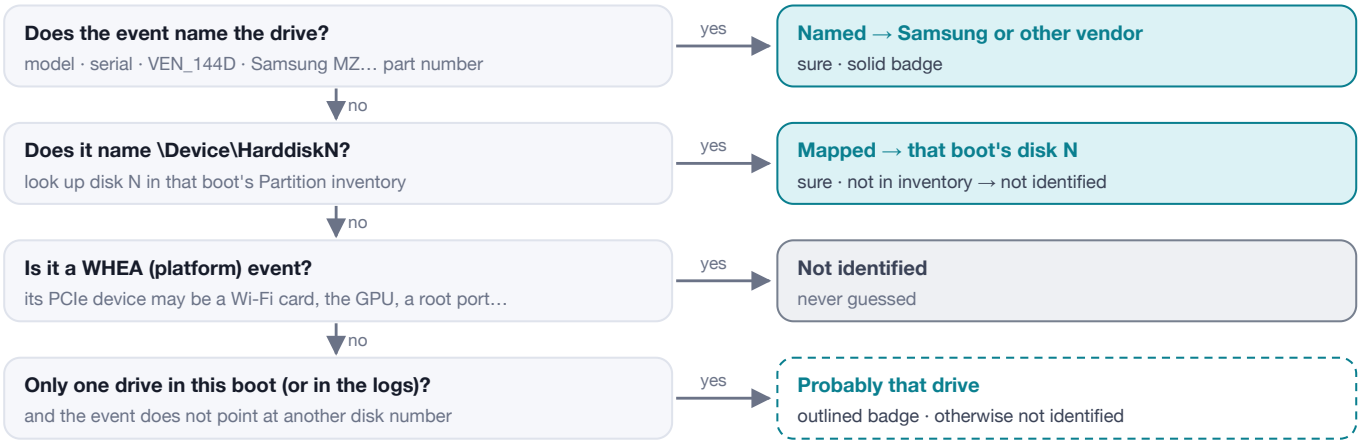
For an **undetermined** power cycle the verdict is weakened one step (Likely → Possibly, No drive evidence → Can't tell). **Effect on the drive** reads the 60 s after boot / wake: drive missing · errors · warnings · healthy · no data. Drive trouble right before a hang or a power-button long-press gets an extra note: it fits a drive that stopped responding.

> **"No drive evidence" is not proof the drive is fine.** Windows keeps its logs on the system drive. If that drive froze suddenly, its last events may never have been written. The app says this next to every such verdict, and recommends checking the drive's own SMART and error logs.

9 Which drive? Samsung identification

> Every storage event is tied to a physical drive — and the app always says how it knows. Samsung's drive problems get their own teal colour everywhere.

Basis	How	Shown as
Named in the event	The event itself names the drive: Storport `VendorId` / `ProductId` / `SerialNumber`, Partition `Model`, a PCI ID such as `VEN_144D` (Samsung) in a WHEA event, or a Samsung part number such as `MZVL2512HCJQ`.	Samsung solid
Disk number	The event names `\Device\HarddiskN`; the Partition/Diagnostic 1006 inventory **of that same boot** says which drive was disk N. Disk numbers can change between boots (a USB drive plugged in, a second disk), so the boot matters.	Samsung solid
Only drive	The event names no drive (e.g. `stornvme 129` on `\Device\RaidPort0`), but only one drive is in that boot — from the boot-time inventory, or else in the whole log set — and the event does not point at a different disk number.	Samsung? outlined = probably
Not identified	Two or more drives and nothing names one; or a WHEA event naming e.g. an Intel root port.	no badge



Making the match reliable

Situation	How WELA handles it
How Samsung is recognised	"Samsung" anywhere in the event · PCI vendor ID `144D` in a device path · a Samsung `MZ…` part number in a model field (OEM drives such as `MZVL2512HCJQ-00B00`).
Storport cuts the name at 16 characters	`SAMSUNG MZVL2512` (Storport) and `SAMSUNG MZVL2512HCJQ-00B00` (Partition) are recognised as one drive by prefix.
Serial in different formats	Case and punctuation are ignored; one log reporting the NVMe EUI-64 (`0025_3881_…`) and the other the real serial still merge when the models match.
Two identical drives	Different serials keep them apart.
USB stick or second disk	Its errors are tied to it by disk number ("Other drive (WDC …)"); with two drives in the boot, unnamed events are not guessed.
No Partition log	It is not read from a folder — add it as a file to map disk numbers. Without it, drives named by Storport are still known; unnamed events are guessed only for Disk 0 and only when one drive is known — and labelled "probably".
Intel / AMD PCI IDs	Deliberately not used: they also name the CPU's own PCIe root ports.

Situation	How WELA handles it
Other vendors	Western Digital/SanDisk, SK hynix, Kioxia/Toshiba, Micron/Crucial, Intel/Solidigm, Seagate, Kingston, Phison, Silicon Motion, ADATA, Lite-On/SSSTC are named so their errors read as "another drive".

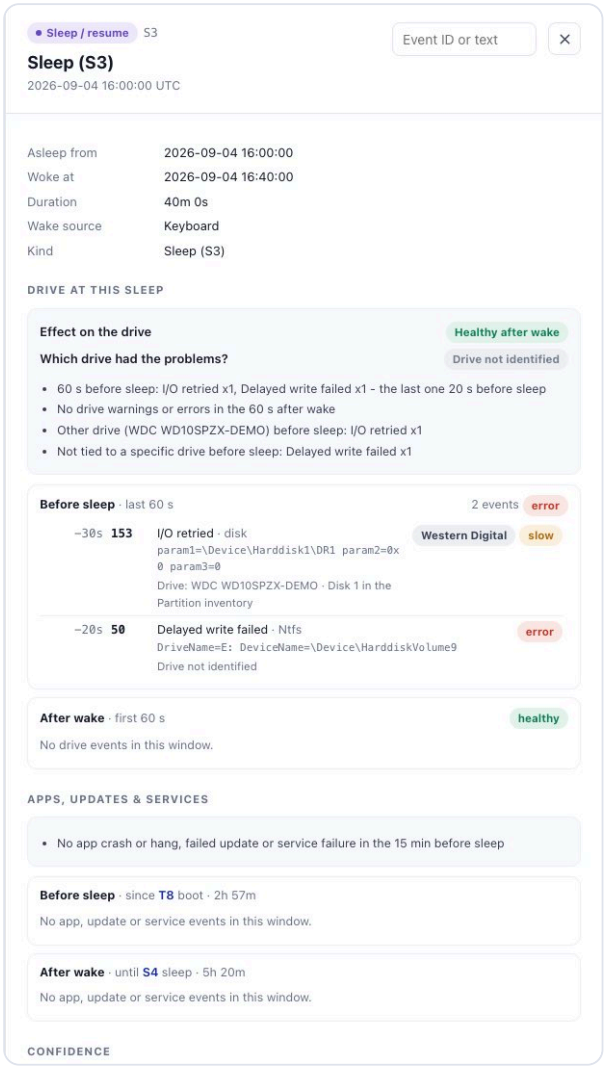
Per power cycle: which drive had the problems?

Samsung drive — at least one warning/error is surely on it · **Probably the Samsung drive** — only by "only drive" · **Another drive** · **Drive not identified** · No drive problems. When the drive is suspected, the reasons spell it out, e.g. "*These drive problems are on the Samsung drive*" or "*…on another drive, not the Samsung drive*", with a breakdown per drive.

Where the highlight appears

- Detail drawer: teal rows and a badge on each event, plus "Drive: <model> · <how identified>"
- Power-cycles table: teal stripe and badge; a **Samsung drive** filter
- Overview: "with Samsung drive problems" tile; badges on suspect power cycles
- Drive tab: Samsung disk card outlined; per-drive counts; drive filter; teal rows
- Timeline: a separate **Samsung drive** lane
- HTML report: teal rows and badges; Excel: teal-filled rows in three sheets

Configurable. The highlighted vendor is one setting (`highlight_vendor`); vendor rules are a list in `catalog.py` .



Demo sleep S3: a USB Western Digital drive's error is pinned on it (Disk 1), and an NTFS error with two drives present is left unidentified — neither is blamed on the Samsung drive.

10 What triggered it? Software

> Windows records who asked for a shutdown (User32 1074), and apps, updates, installers and services leave their own trail — WELA reads both.

Sources

Source	Events (log)	Tells us
Shutdown request	User32 1074 (System)	Which process asked, for which user, why (reason code), restart or power off
Windows Update	WindowsUpdateClient 19, 20, 21, 22, 43 (System); Servicing 2, 4 (Setup)	Installed / failed / needs a restart / restart scheduled / install started
Installers	MsiInstaller 1033–1038, 1005, 1040/1042, 11707/11708/11724 (Application)	Installed / removed / needs a restart / **restarted the PC** / transaction open
Apps	Application Error 1000, Application Hang 1002, .NET Runtime 1026, Windows Error Reporting 1001 (Application)	Crashes, hangs; WER reports incl. **BlueScreen** and LiveKernelEvent
Services	Service Control Manager 7031, 7034, 7023, 7024, 7043, 7022, 7011 (System)	Crashes (and whether recovery restarts the PC), services holding up shutdown, timeouts
System	Display 4101, Resource-Exhaustion-Detector 2004, Kernel-Power 105	Display driver timeouts, low memory, AC ↔ battery

Anatomy of User32 1074

Field	Example
1 · process	`C:\Windows\servicing\TrustedInstaller.exe (PC)`
3 · reason text	Operating System: Upgrade (Planned) — *localised*
4 · reason code	`0x80020003` — *same in every language*
5 · type	restart / power off
6 · comment	"The system process … lsass.exe terminated unexpectedly…"
7 · user	NT AUTHORITY\SYSTEM

Decoding the reason code

Bit 31 = planned · bits 16–23 = major · bits 0–15 = minor (Windows `SHTDN_REASON_*` constants).

Code	Decoded	Category
`0x80020003`	Operating System: Upgrade (planned)	Windows Update
`0x80020010`	Operating System: Service pack (planned)	Windows Update
`0x80040002`	Application: Installation (planned)	Installer
`0x10007`	Hardware: Disk (unplanned)	Hardware reason
`0x50006`	System: Unstable (unplanned)	System unstable
`0x0`	Other (unplanned)	— (typical user shutdown)

Who asked? Initiator categories

Category	Example processes (from the 1074 path)
Windows Update	TrustedInstaller, TiWorker, MoUsoCoreWorker, UsoClient, wuauclt, MusNotification, SetupHost, SIHClient — *or any process with an OS-servicing reason code*
OEM / vendor tool	Dell Command Update (dcu-cli), SupportAssist, Lenovo, HP, **Samsung Magician**, firmware / BIOS updaters

Category	Example processes (from the 1074 path)
IT management	SCCM (CcmExec), Intune (OmaDmClient, IntuneManagementExtension, AgentExecutor), Group Policy scripts, WMI
Software installer	msiexec, *setup*, *install*, *uninst*
Command or script	shutdown.exe, PowerShell, cmd, wscript / cscript
System process crash	wininit / lsass / csrss with an "unstable" reason or a "terminated unexpectedly" comment
User request	Explorer, RuntimeBroker, StartMenuExperienceHost, ShellExperienceHost, Winlogon, LogonUI

When the process and the reason disagree, the reason code wins for updates: "Update and restart" from the Start menu is requested by the user's shell, but the reason code says OS servicing — so it counts as Windows Update.

The verdict rules

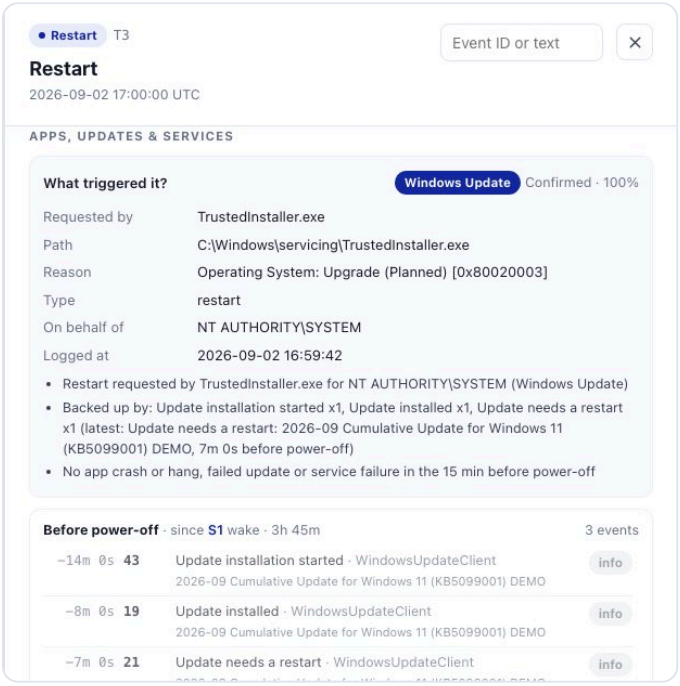
Evidence	Verdict	Confidence
1074 names the process that asked (any power cycle)	**Confirmed** + category	0.70; +0.15 if the reason code agrees; +0.15 if backed by update / installer / service events
Orderly stop, no 1074 — but MsiInstaller 1005, Windows Update 22, or a 7031 service crash whose recovery restarts the PC	**Likely**	0.60
Restart, no 1074 — an update or install was waiting for a restart (WU 21, MSI 1038, Servicing 4)	**Possibly**	0.35
Orderly stop, nothing of the above	Not logged	— (e.g. short power-button press)
Hard shutdown after a system process crash (csrss, lsass, wininit …)	Likely with crash code 0xEF / 0xF4, else Possibly	0.50 (+0.20 with the crash code)
Hard shutdown after a service crash whose recovery restarts the PC (7031) — the restart did not complete	**Likely**	0.60
Hard shutdown while an update (WU 43) or install (MSI 1040) was still running	**Possibly**	0.40
Hard shutdown right after apps / services hung (1002, 7043, 7011)	**Possibly**	0.35
Hard shutdown after a display-driver timeout (4101) / an app crash (1000)	**Possibly**	0.30 / 0.25
Logs cover the window and nothing was found	No software evidence	—
No log covers the window, or the power cycle is undetermined	Can't tell	—

Symptom, not cause. If the drive *also* logged problems right before power-off, the software verdict says so and loses 0.15: a drive that stalls makes apps and services hang or crash, so those may be a symptom of the drive. (A crash code that both verdicts read does not trigger this — only real drive events do.)

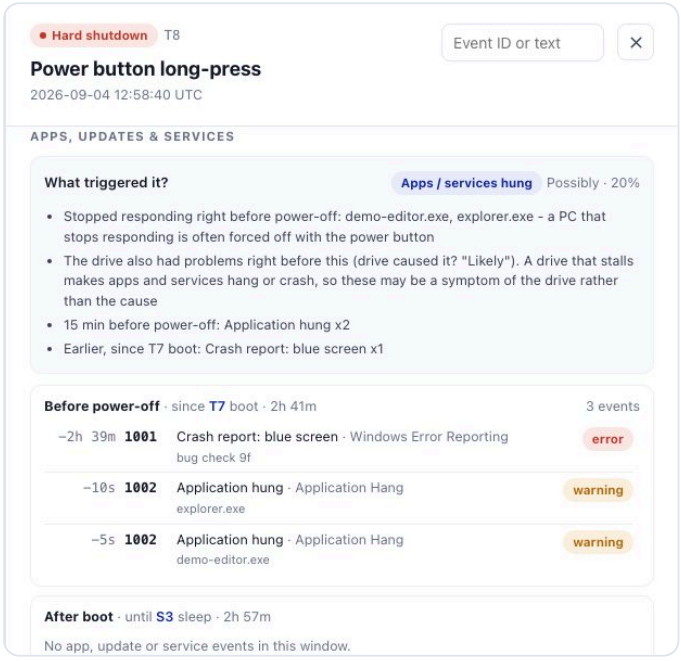
Also reported as context: an update waiting for a restart since earlier in the session, the PC running on battery (Kernel-Power 105), a blue-screen report or an update finishing after the next boot, and services that held up a hung shutdown (7043).

Examples from the demo case

ID	Power cycle	Trigger	Why
T3	Restart	**Windows Update** Confirmed · 100 %	1074 by TrustedInstaller.exe, reason 0x80020003 (OS upgrade, planned); backed by WU 43 → 19 → 21 for KB5099001
T16	Restart	**System process crash** Confirmed · 100 %	1074 by wininit.exe, reason 0x50006 (System: unstable), comment "lsass.exe terminated unexpectedly"; Application Error for lsass.exe 20 s earlier
T13	Restart (no 1074)	**Software installer** Likely · 60 %	MsiInstaller 1005 "restarted the PC" for DEMO Tools 1 min before power-off
T15	Hard: power loss	**Update was installing** Possibly · 40 %	WU 43 started KB5099002 and never finished — it was the last thing logged
T8	Hard: long-press	**Apps / services hung** Possibly · 20 %	explorer.exe and an editor hung seconds before — but the Samsung drive was resetting at the same time, so flagged as a likely symptom; the list also goes back to the blue-screen report from T7's boot, 2 h 39 min earlier
T2	Shutdown	**User request** Confirmed · 85 %	1074 by RuntimeBroker.exe (Start menu) for the user, reason 0x0



T3 — Windows Update restart: requester, path, reason code, user, the update events before it.



T8 — app hangs before a long-press, flagged as a possible symptom because the drive was resetting at the same time. The list runs back to T7's boot; the verdict sums that up as "Earlier".

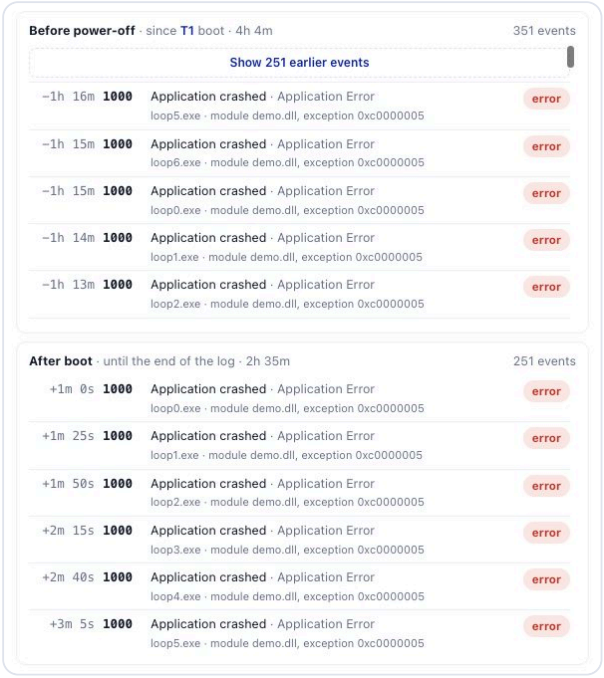
The lists: from one power cycle to the next

The verdict reads the minutes right around the power cycle, but a cause can start much earlier — an app that has been hanging for an hour, an install that began in the morning. So the app / update lists shown with each power cycle are not cut at 15 minutes:

List	From	To	Opens at
Before power-off	the boot / wake that ended the power cycle before (or the start of the log)	this power-off / sleep	the bottom — the last events before power-off; scroll up for older ones
After boot	this boot / wake	the power-off / sleep of the next power cycle (or the end of the log)	the top — the first events after boot

- One power cycle's "after" list is the next one's "before" list — the same stretch, opened from the other end.
- A sleep that never resumed ends no stretch: the hard shutdown after it lists everything since the wake before.
- Problems older than the verdict window are summed up in the reasons, e.g. "*Earlier, since T7 boot: Crash report: blue screen x1*" (T8).
- The search box at the top of the drawer finds an event ID in every list of the power cycle, including events not loaded yet.

A count cap, not a time cap. Each list loads the 100 events nearest the power cycle (`software_list_max`), however old; **Show … earlier / later events** loads the rest of the stretch from the case's saved events (up to 5,000), and the rows on screen stay where they are. A time cap — say the last hour — would hide exactly the slow causes these lists are for; a count cap keeps the drawer fast even in a crash loop of thousands of events, and nothing is out of reach.



A busy case: 351 app crashes before a hard shutdown and 251 after it. The "before" list is scrolled up to its top — 100 loaded, one click for the other 251; the "after" list as it opens, at the first event after boot.

11 Data-quality checks

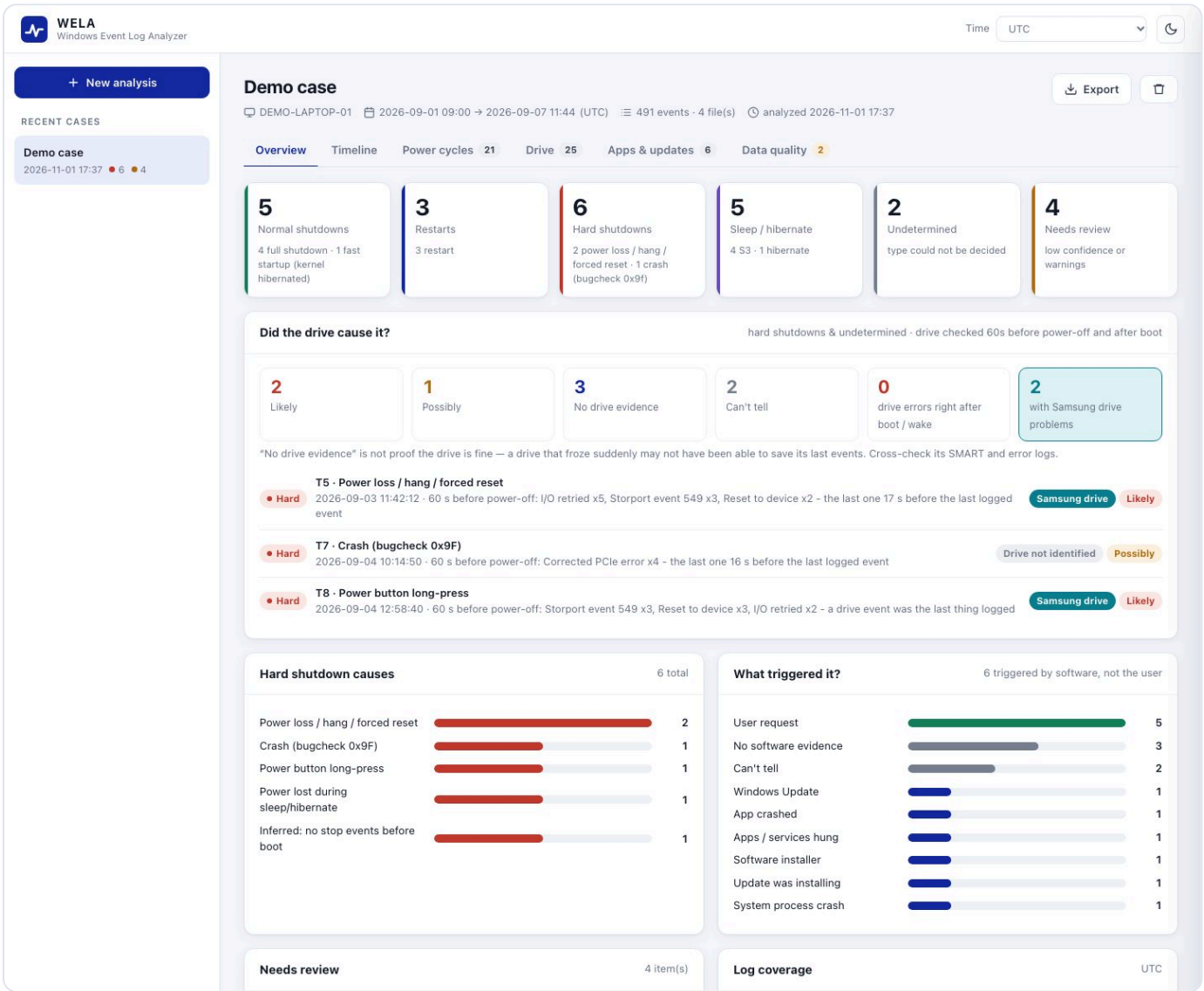
> Before trusting any verdict, WELA checks the logs themselves — and lowers confidence or refuses to guess where they are incomplete.

Check	How	Effect
Missing logs	System / Application / Storport expected	Warning; affected verdicts say which log was missing
Unreadable files	Each file read separately; errors captured	Listed with the reason; analysis continues
Missing records	Gaps in record IDs per file and log	Power cycles at a gap → undetermined or −0.30 confidence
Cleared logs	Eventlog 104 / 1102	Warning with time and who cleared it

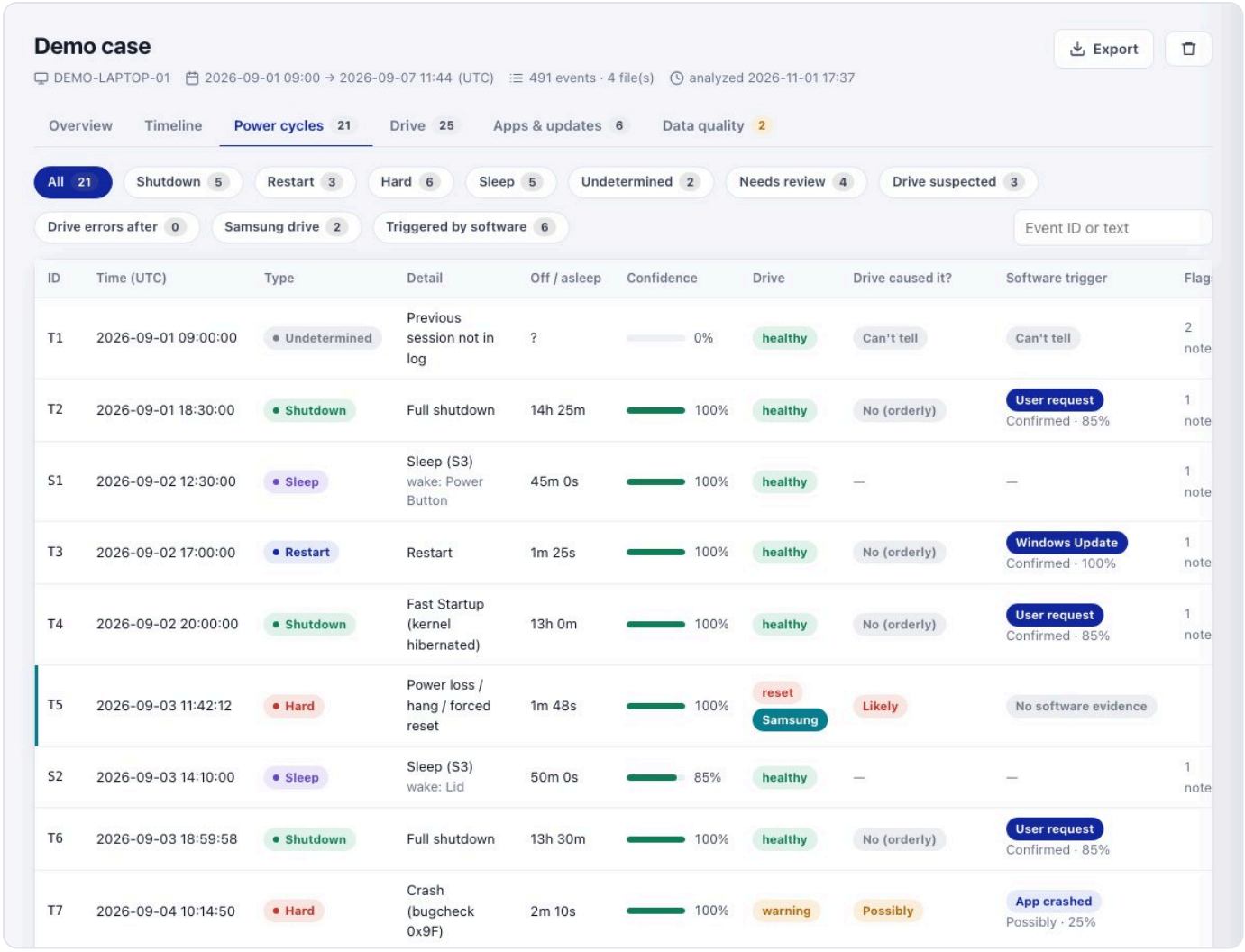
Check	How	Effect
Clock changes	Kernel-General 1 with a jump ≥ 5 min (not during boot)	Listed with the process that changed the clock
Time going backwards	Record order vs. timestamps (> 60 s)	Warning
Unexplained quiet gaps	≥ 2 h with no events inside a session and no sleep	Info: possibly an unlogged sleep — or an idle PC
Storport coverage	Is there Storport data for each drive window?	Note on each verdict where only System-log disk events were checked
Logs skipped in a folder	Only System, Application and Storport are read from a folder	Count shown after the analysis and under Data quality → Files

12 Tour of the application

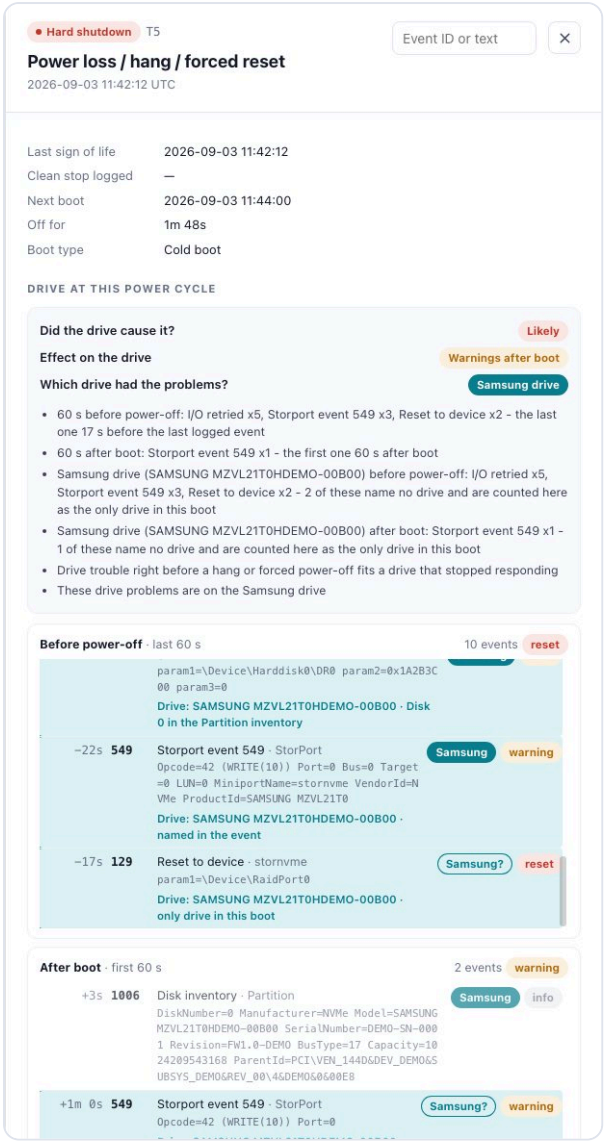
Six tabs per case. Start the app with `python run_wela.py`; the browser opens at http://127.0.0.1:8765.



Overview. Tiles per power-cycle type (click to filter). "Did the drive cause it?" with the teal *with Samsung drive problems* tile and Samsung badges on the suspect power cycles. "What triggered it?" breaks power cycles down by software trigger (click a bar to list them). Hard-shutdown causes, needs-review list, log coverage, storage and app/update summaries follow below.



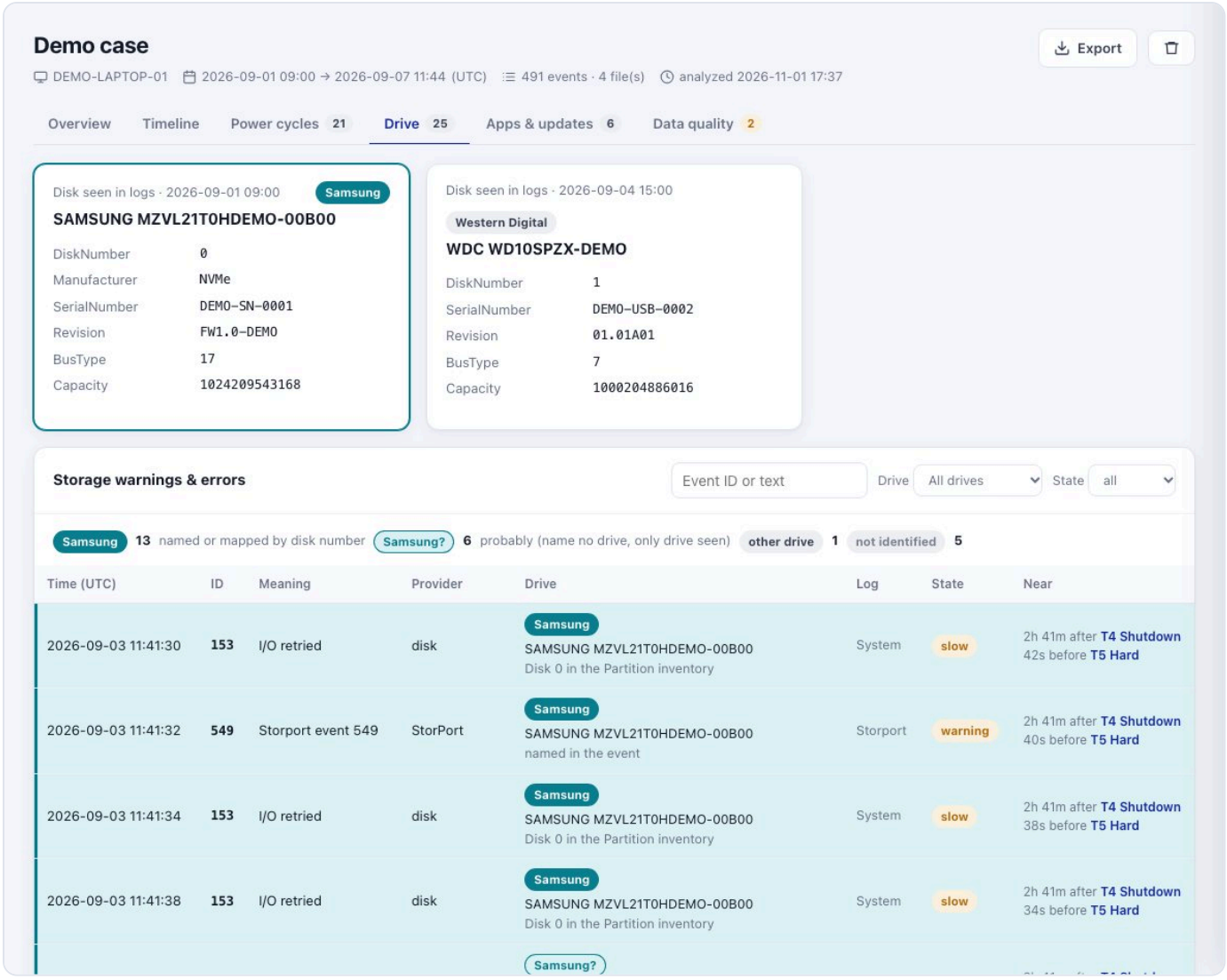
Power cycles. One row per power cycle and sleep: time, type, detail, off time, confidence, drive state with Samsung badge (teal stripe on the row), drive verdict, software trigger with its strength and confidence, flags. Filters include *Samsung drive* and *Triggered by software*. Search: a number finds those event IDs exactly — in the evidence and in each power cycle's drive and app lists; anything else searches the text.



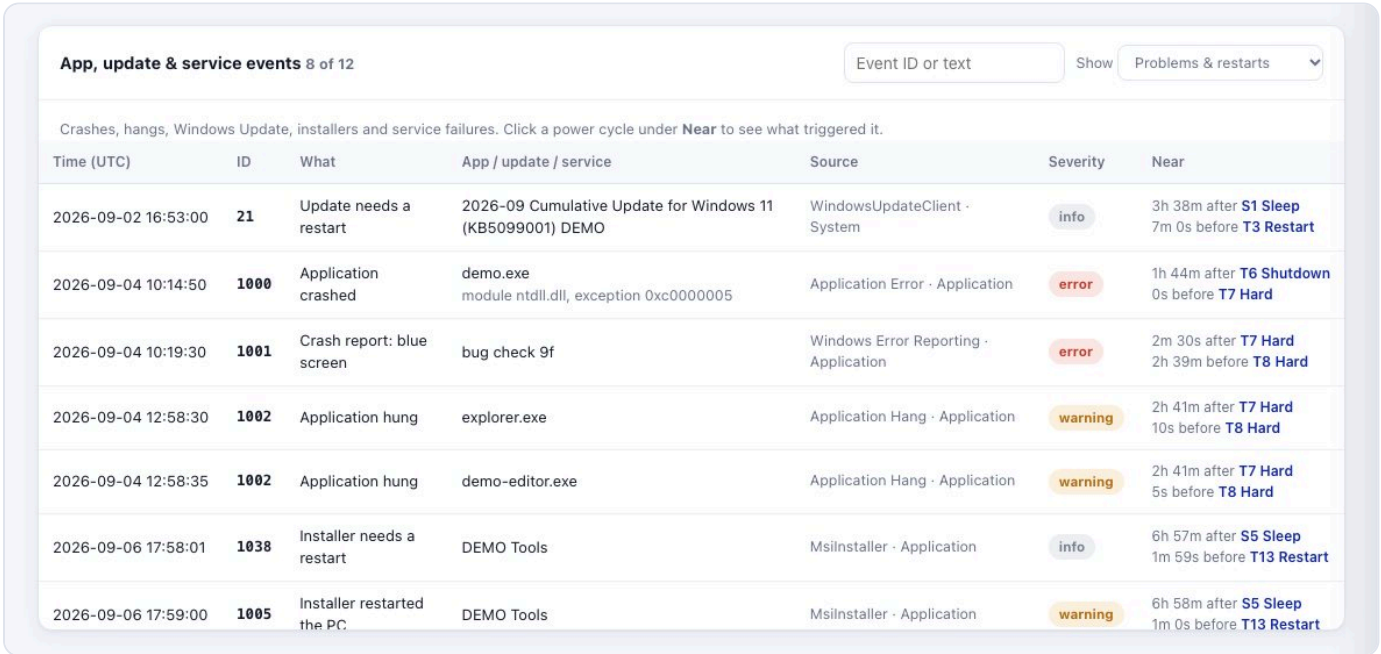
Detail drawer — the evidence for one power cycle (T5)

- **Key facts** — last sign of life, clean stop, next boot, off time, boot type.
- **Drive** — "Did the drive cause it?" (Likely), "Effect on the drive", and **"Which drive had the problems?"** (Samsung drive), with plain-language reasons.
- **Drive windows** — every storage event 60 s before power-off and after boot with offsets and decoded fields; Samsung rows tinted teal; each row says how the drive was identified ("named in the event", "Disk 0 in the Partition inventory", "only drive in this boot"). Each list is a scroll box that opens at the end nearest the power cycle.
- **Apps, updates & services** — "What triggered it?", the 1074 details, and the app / update lists from the power cycle before to the one after (chapter 10).
- **Confidence, flags, evidence found / missing** — each event has a *context* button showing the raw events around it.
- **Search** (top right) — an event ID or text, across every list in the drawer.

Point out in the demo: retries, Storport warnings and two resets (32 s and 17 s before power-off). The Storport warnings are *named in the event*, the retries are mapped through *Disk 0*, and the resets — which name no drive — show as Samsung? : the app is honest about which links are certain.



Drive tab. Disk cards from the inventory (Samsung outlined in teal, other vendors labelled), a per-drive summary (named or mapped / probably / other drive / not identified), drive and state filters, and every storage warning/error with the drive and how it was identified. **Near** shows the time since the power cycle before and to the one after (click either to open it); search by event ID or text.



Apps & updates tab. Every crash, hang, Windows Update, installer and service event, filterable by kind and searchable by event ID, with the app / update / service named; **Near** gives the time since the power cycle before and to the one after.



Timeline. Zoom and pan across the whole log period: powered-on sessions, transitions, sleeps, a dedicated **Samsung drive** lane, other storage events, the **Apps & updates** lane, and log coverage per channel; record gaps shaded. Click any marker for details.

Also: **Data quality** tab (chapter 11), a case list in the sidebar (cases are saved locally), time-zone selector, dark mode, and analysis settings on the home screen (drive window, software trigger window, restart gap, review threshold, quiet-gap length). The path box on the home screen starts empty for every new analysis.

13 Exports and command line

> Everything on screen can leave the app in a form that fits the claim: a readable report, a filterable workbook, or raw data.

Formats

Format	Use
HTML report	One self-contained file to read or share: summary tiles, every power cycle with drive and software verdicts, the drive events around each power cycle (Samsung rows highlighted), the software verdicts and the app / update problems around each power cycle, coverage and data-quality notes.
Excel	Attach to the claim; filter and sort (sheets below).
CSV	One row per power cycle, UTF-8 with BOM for Excel.
JSON	The full result, for tooling and future AI features.

Excel workbook — 7 sheets

Sheet	Contents
Summary	Counts, drive verdicts, Samsung counts, software triggers
Power cycles	All verdicts and reasons (22 columns); Samsung rows filled
Drive windows	Every storage event around every power cycle, with drive and how identified; Samsung warnings / errors filled teal
Software windows	The 100 app / update / service events nearest each power cycle, before and after
Storage events	All storage warnings / errors with drive attribution
Software events	All app / update / service events
Coverage	Each log's time range and event count

Times are converted to the time zone chosen in the UI (e.g. IST). Frozen header row and auto-filter on every sheet.

Command line

```
# start the UI (default) — or press F5 in VS Code and pick "WELA: start UI"
python run_wela.py
# analyse without the UI and write a report (.html, .xlsx, .csv or .json); --tz is minutes from UTC (330
= IST)
# from a folder only System, Application and Storport are read; the other logs are counted as skipped
python -m wela analyze "D:\Claims\DELL-123\Logs" --out report.xlsx --tz 330
# write the demo logs to a folder
python -m wela demo demo_logs
# run the tests
python -m unittest discover -s tests -v
```

The CLI prints one line per power cycle, e.g.

```
  Samsung drive: suspected in 2, problems around 2 power cycle(s), 19 warning/error event(s)   triggered
by software: 6
    T5  2026-09-03T11:42:12  hard      Power loss / hang / forced reset   100%  drive:reset   caused it?
Likely [Samsung]  trigger: No software evidence
    T8  2026-09-04T12:58:40  hard      Power button long-press            100%  drive:reset   caused it?
Likely [Samsung]  trigger: Apps / services hung (Possibly)
```

14 Quality, testing and performance

> 51 automated tests pin down every rule; a demo case with known answers checks the whole pipeline; a random-input test checks it never crashes.

Test group	Tests	What it proves
Demo scenario	10	All 4 demo logs parse; all 16 power cycles and 5 sleeps get the expected verdict; drive cause/effect, Samsung attribution, software triggers, the app / update lists from one power cycle to the next, data-quality findings and all four exports are as expected.
Power-cycle rules	13	Clean shutdown with 6006 missing, restart inferred from gap, Fast Startup, every 41 subtype, hang during shutdown, silent end with low confidence, sleep rebuilt from Power-Troubleshooter 1 only, drive verdicts, XML parsing.
Which drive	10	Storport naming, Samsung part number without the name, other vendors, disk-number mapping, **disk numbers that swap between boots**, "only drive" inference, no-inventory rule, WHEA never guessed, serial formats merged, configurable vendor.
Software trigger	15	Reason-code decoding, user / Windows Update / "Update and restart" / vendor tools / installer, service-recovery restart, no-1074 cases, system-process crash, hang as symptom, update still installing, shutdown requested but hung, battery note, named and unnamed event fields; lists keep the nearest events and sum up older problems.
Folder selection	2	Only System, Application and Storport (Operational, Health) are read from a folder — by winevt\Logs and by exported file names; other logs are skipped and counted; a file given by itself is always read; the demo folder skips its Partition log.
Robustness	1	30 rounds of up to 400 random, malformed events (bad times, garbage fields, huge numbers, odd Unicode) through the engine and every export — nothing may crash.

The demo case — 16 power cycles and 5 sleeps, each a known tricky case

ID	Scenario	ID	Scenario
T1	Log starts at a boot → undetermined	T9	Power lost while asleep
T2	Normal shutdown by the user	T10	Session just ends → inferred, 30 %

ID	Scenario	ID	Scenario
T3	Windows Update restart	T11	37 records missing → undetermined
T4	Fast Startup shutdown	T12/T14	Normal shutdowns
T5	Power loss after Samsung drive resets	T13	Restart without 1074 — installer restarted the PC
T6	Clean shutdown with 6006 missing	T15	Power lost while an update was installing
T7	Crash 0x9F with WHEA errors and an app crash; blue-screen report after boot	T16	lsass.exe crash → Windows restarts
T8	Long-press while the Samsung drive resets and apps hang	S1–S5	Full sleep, entry missing, Power-Troubleshooter only (with USB-drive errors), never resumed, hibernate

Other checks done for this release

- Every tab and the detail drawer rendered in a real browser (headless Chrome) — the demo case, a case saved by the previous version and a busy case — with a check for script errors.
- A case saved by the previous version opens without errors; its app / update lists keep their old 15 / 10-minute form until it is re-run.
- A busy case — crash loops of 351 and 251 events around one hard shutdown: each list opens at the 100 nearest, *Show more* loads the rest without moving the rows in view, and the drawer search finds events that were not loaded yet.
- A folder holding Setup, Security and Partition logs next to the four: 3 skipped; Storport Health read and shown as its own log.
- End-to-end through the real local server: analysis of a folder, the demo, every export, the event API, delete.
- Python 3.9 compatibility (the oldest supported version) — the suite runs on 3.9.

Performance (analysis step, measured on a laptop)

0.2 s
19,640 events · 640 power cycles

1.7 s
98,200 events · 3,200 power cycles

4.7 s
196,400 events · 6,400 power cycles

Reading `.evtx` with wevtutil on Windows adds time proportional to file size. Time grows somewhat faster than the event count (10× the events took about 20× the time) but stays in seconds for very large logs.

15 Configuration and tuning

> All knowledge lives in `wela/catalog.py`; a `wela_config.json` next to `run_wela.py` overrides any part of it without code changes.

Section	Contents
signals / signal_labels	Provider + event ID for every power signal (boot, stop, unclean, sleep …) and its display label
weights	Evidence weights and penalties for power-cycle confidence; `weights.software` for "what triggered it?"
storage_events	Storage event IDs with label and state (missing / reset / error / slow / warning / info); `storage_provider_hints`, `platform_provider_hints`
storage_bugchecks	Crash codes that point at storage, "strong" or "possible"
drive_vendors / highlight_vendor	Vendor rules (names, PCI IDs, part-number patterns) and the vendor to highlight (Samsung)
software_events	App / update / install / service / system event IDs with label, group and kind
shutdown_initiators	Process-path patterns → initiator category (first match wins)

Section	Contents
critical_processes / critical_bugchecks	csrss, wininit, winlogon, lsass, smss, services · 0xEF, 0xF4
settings	Time windows and thresholds (below)
restart_words / poweroff_words	1074 shutdown-type words in many languages (English, German, French, Spanish, Italian, Dutch, Japanese, Chinese, Korean, Russian)

Setting	Default
boot_cluster_s — boot markers counted as one boot	180 s
boot_window_s — look for 41/6008/1001 after boot	900 s
tail_window_s — look for stop signals before the end	1,800 s
restart_gap_s — stop → boot gap = restart	300 s
long_restart_gap_s — restart taking too long	1,800 s
pt1_merge_s — sleep report joins a nearby resume	300 s
drive_window_s — drive window before / after	60 s

Setting	Default
software_window_s — trigger verdict reads before	900 s
software_after_s — … and after boot	600 s
software_list_max — app / update events per list	100
running_lookback_s — update / install still running	3,600 s
unexplained_gap_s — quiet gap worth flagging	7,200 s
time_change_flag_s — clock jump flagged	300 s
review_threshold — confidence for "needs review"	0.6

```
// wela_config.json — only the keys you want to change
{ "settings": { "drive_window_s": 120, "software_window_s": 1800 },
  "highlight_vendor": "Samsung" }
```

The most-used settings are also on the home screen under *Analysis settings*, per analysis. Which logs a folder gives is decided in `wela/parser.py` (`FOLDER_LOGS`, `folder_log`).

16 Limitations — and what to say

> Stating the limits up front builds trust: WELA is precise about what the logs show and explicit about what they cannot show.

Limitation	What to say / mitigation
Validated on synthetic logs so far	Every rule is tested against a demo case with known answers. Next step: run on 20–30 closed claims with known outcomes and on lab PCs where each power-cycle type is triggered on purpose, then tune the catalog.
Storport field names and a few event IDs confirmed only in the demo	WELA reads the standard Storport fields and also scans every field for "Samsung", PCI 144D and MZ… part numbers. First real-claim check: open a Storport event in the drawer and confirm the "Drive" line says *named in the event*. Lab-test Windows Update 21/22, Servicing 4 and the 7031 recovery action.
The logs live on the drive under suspicion	A drive that froze may never have saved its last events, so "No drive evidence" is not "healthy". Cross-check with the drive's own SMART and error logs.
"Probably" drive attributions	Unnamed events are only attributed when one drive exists, and are labelled. Add the Partition log — as a file, it is not read from a folder — to turn more of them into certain matches.
Who asked for a shutdown is not always logged	No 1074 means "Not logged"; restart vs. shutdown then falls back to the stop → boot gap (5 min), flagged as inferred.
Rules, not judgement	The engine reports evidence and patterns; it does not replace failure analysis of the drive. Every verdict can be reviewed and overridden by the engineer.

Limitation	What to say / mitigation
One PC per case	No cross-claim view yet (roadmap: trends by model / firmware).
Windows event logs only	No minidumps, drive telemetry or SMART import yet (roadmap).
Folder logs are picked by file name	A log exported under an unrelated name (e.g. `logs1.evtx`) is skipped when it sits in a folder — add it as a file. The number of skipped logs is shown after every analysis.
Display caps	Each drive window lists up to 300 events. Each app / update list shows the 100 nearest the power cycle and loads the rest of the stretch on request, up to 5,000. Case-wide lists hold up to 5,000 (problems kept first). Counts always cover everything.

17 Demo script and likely questions

> About 15 minutes: problem → one click → four questions answered with evidence → exports → what's next.

Before the meeting. Check out branch `samsung-drive-and-app-triggers` · run `python run_wela.py` · set Time to **IST** · click **Try demo data** once to warm up · have a real (closed) claim's logs ready as a backup · keep this document open on chapter 16.

0:00 **1 min**	**The problem.** "When an SSD comes back after crashes or sudden shutdowns, we need to know what happened to the PC, whether our drive was involved, and whether something else — Windows Update, an app, the user — triggered it. That evidence is in the Windows logs, but reading them by hand takes hours."
1:00 **1 min**	**One click.** Home → *Try demo data*. **Say:** "Four logs, 491 events, analysed in under a second, on this PC — nothing installed, nothing sent anywhere."
2:00 **2 min**	**Overview.** Tiles: 5 shutdowns, 3 restarts, 6 hard shutdowns. "Did the drive cause it?": 2 likely, 1 possibly — and the teal tile: 2 power cycles with **Samsung drive** problems. "What triggered it?": Windows Update, installer, system-process crash… **Say:** "Every number is clickable and every verdict shows its evidence."
4:00 **3 min**	**T5 — the drive case.** Power cycles → T5. Drive resets and retries in the last minute before power-off, all on the Samsung drive. Point at how each was identified: Storport warnings *named in the event*, disk retries via *Disk 0 in the Partition inventory*, and the resets (which name no drive) as Samsung?. **Say:** "It tells us which links are certain and which are probable."
7:00 **2 min**	**T8 — symptom vs. cause.** Power-button long-press; Samsung drive resetting; apps hung seconds before. Scroll to *Apps, updates & services*: "Apps / services hung — possibly", with the note that hangs are probably a symptom of the drive. The list opens at the hangs seconds before power-off; scroll up and it goes back to T7's boot — the blue-screen report 2 h 39 min earlier. **Say:** "It doesn't blame the software just because apps hung — and it doesn't cut the story at 15 minutes."
9:00 **2 min**	**Software triggers.** T3 — Windows Update, confirmed: TrustedInstaller, reason code 0x80020003, the update events before it. T16 — lsass.exe crashed, Windows restarted itself. T15 — power lost while an update was installing. **Say:** "These would otherwise look like 'random restarts' in a claim."
11:00 **1 min**	**Not our drive.** S3 — a USB Western Digital drive's errors are pinned on it, not on the Samsung SSD. Drive tab → filter *Samsung drive*; type `129` in the search box to list every reset.
12:00 **1 min**	**Timeline.** The Samsung lane and the Apps & updates lane line up against the hard shutdowns.
13:00 **1 min**	**Exports.** Export → Excel: Samsung rows filled teal in *Drive windows*; HTML report for sharing.
14:00 **1 min**	**Close.** Validation on closed claims next, then the AI layer: plain-language claim summaries and "ask the case" — on top of the rules, never replacing them (chapters 18–19).

Likely questions

Question	Answer
How accurate is it?	The rules use standard Windows event IDs and are pinned by 51 tests. Accuracy on real claims will be measured next on 20–30 closed claims with known outcomes. Meanwhile, confidence scores and "needs review" flags show where it is unsure.
Can it prove the drive is fine?	No — and it says so. The logs are stored on the drive itself; a sudden freeze can lose the last events. It points to SMART and drive-side logs for that.

Question	Answer
How does it know it's our drive?	Storport events usually name the drive; disk events are mapped through the boot's disk inventory; anything else is "probably" (only one drive) or "not identified". The basis is shown on every event.
What if the PC has two drives or a USB stick?	Disk numbers are mapped per boot; unnamed events are not guessed when there are two drives (demo S3).
Is customer data safe?	Everything runs on the engineer's PC; the server only listens on 127.0.0.1; there are no external calls.
How long does it take?	About 4.7 s for ~200,000 events (analysis), plus reading the files.
Which logs do we need from the customer?	System (required); Application and Storport (strongly recommended) — or simply a copy of the whole `winevt\Logs` folder: WELA reads only those from a folder. Partition (optional) as a file of its own. One `wevtutil` command each — a collection script is on the roadmap.
Why not let AI decide?	Claim decisions must be reproducible and auditable. Rules give the same answer every time with the evidence. AI is best used on top: summarising, answering questions, triaging, spotting patterns.
Can the rules be changed?	Yes — all in `catalog.py`, or a JSON override file; no engine changes.
What does it take to run?	Python 3.9+ on Windows; no packages. ~4,900 lines of code, standard library only.

18 Roadmap

> First prove it on real claims; then widen the evidence (drive-side data, dumps); then scale from one PC to the whole claim population.

Item	Why	Effort
Now — validate and harden		
Validate on 20–30 closed claims	Measure agreement with known outcomes; tune weights and thresholds; confirm Storport field names on real logs	M
Lab validation	Trigger each power-cycle type on purpose (update restart, installer restart, long-press, power pull, crash) and check the verdicts	S
One-click log collection	A signed script for customers / field engineers that exports the logs WELA reads (plus Partition) into one zip	S
Windows package	Single executable so engineers do not need Python	S
Next — more evidence		
Drive-side data import	NVMe SMART / health log and error log: automatic power-cycle and unsafe-shutdown cross-check, media errors next to each power cycle	M
Crash-dump summary	Read minidump bugcheck parameters and the faulting driver next to the crash verdict	M
Drive-tool and OEM logs	Firmware-update history and tool events (e.g. Samsung Magician, OEM updaters) on the timeline	M
Later — scale		
Batch mode	Analyse a folder of claims at once; one summary row per claim	S

Item	Why	Effort
Claims dashboard	Trends by drive model, firmware, OEM platform and Windows build (e.g. reset storms after a firmware release)	L
Claim-system integration	Attach WELA results to the claim record automatically through an API	M
AI layer	Chapter 19	S → L

Effort: S = days, M = a few weeks, L = a quarter-scale project — rough sizes for planning, to be refined.

19 AI integration

> Rules decide; AI explains. The deterministic engine stays the source of truth for verdicts — AI turns its results into summaries, answers and patterns.

Why this split

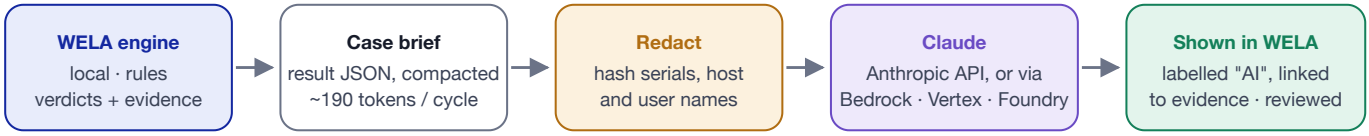
What rules are best at. Same input → same verdict, every time; the evidence behind each verdict is explicit; easy to audit and to defend in a claim decision.

What AI (large language models) is best at. Reading and summarising large, messy text; explaining technical evidence in plain language or another language; answering open questions; spotting patterns across many cases.

Use cases

Use case	What it does	How	Effort
1 · Claim summary writer	Writes the plain-language case summary for the claim record — what happened, when, whether the Samsung drive was involved — citing power-cycle IDs and events; optionally in the customer's language.	One LLM call on WELA's result (not the raw logs)	S
2 · "Ask the case"	Chat in the UI: "What happened in the 10 minutes before the crash on 4 September?" — the model looks up events through WELA's own local API and answers with links to the evidence.	LLM with tool use calling WELA's endpoints	M
3 · Unknown-event explainer	Explains events the catalog does not know yet (new Storport or driver IDs) and drafts catalog entries for an engineer to approve.	LLM on the unknown events + catalog	M
4 · Claim triage at scale	Sorts incoming claims into buckets — drive suspected, software-triggered, power / user, insufficient logs — to prioritise engineering time.	Small, fast model over batch results	M
5 · Confidence calibration	Learns the evidence weights from closed claims with known outcomes, so "80 %" really means 80 %.	Classical machine learning (e.g. logistic regression) — not an LLM	M
6 · Fleet patterns	Finds recurring patterns across many claims by model, firmware and platform, and describes them for engineers.	Statistics + LLM descriptions	L
7 · Crash-dump assistant	Summarises debugger output for a crash dump next to the WELA crash verdict.	LLM on debugger text	M

Architecture for the AI layer



Opt-in per case · AI never changes a verdict · the engineer approves what goes into the claim

Guardrails

- **Verdicts stay rule-based.** The model may explain, summarise and suggest — it never edits a verdict, a confidence or the evidence.
- **Evidence-linked output.** Every statement cites power-cycle IDs or event IDs the engineer can click; unsupported claims are flagged.
- **Data minimisation.** Send the compacted result, not raw logs; hash serial numbers, host names and user names; opt-in per case.
- **Approved deployment.** Use the provider route the company already has an agreement with (Anthropic's API directly, or Claude through Amazon Bedrock, Google Vertex AI or Microsoft Foundry), with data-retention terms reviewed by IT and legal.
- **Measured before rollout.** Build an evaluation set from closed claims and score summaries for correctness before engineers rely on them; keep a log of prompts and outputs for audit.
- **Optional module.** The core app stays dependency-free; the AI layer is a separate add-on that needs the `anthropic` package and network access.

Model choice and cost (list prices per million tokens, October 2026)

Model	Input / output	Fit
Claude Opus 5.5	$4 / $20	Default for claim summaries, "ask the case" and the unknown-event explainer — best reasoning on technical evidence
Claude Sonnet 5.5	$2 / $10	High-volume summaries if cost matters more
Claude Haiku 5.5	$0.10 / $0.50	Triage of many claims, classification
Batch API	50 % off	Overnight processing of many claims (not interactive)

Estimate. A compacted brief is about 190 tokens per power cycle (measured on the demo case). A claim with 100 power cycles ≈ 20,000 input tokens (≈ $0.08 on Claude Opus 5.5) plus a few thousand output tokens including the model's reasoning (≈ $0.04–0.08): roughly **10–20 US cents per claim summary**, about half with the Batch API, and well under one cent per claim for triage with Claude Haiku 5.5. Estimates from list prices; measure on real claims.

What a first integration looks like (claim summary writer)

```
import json, anthropic

client = anthropic.Anthropic()                      # credentials from the environment
case = json.load(open("cases/<case-id>/result.json"))
brief = {k: case[k] for k in ("computer", "coverage", "counts", "hard_breakdown", "software_breakdown")}
brief["items"] = [{"id": i["id"], "type": i["type"], "subtype": i["subtype"], "confidence":
i["confidence"],
                   "drive": {k: i["drive"].get(k) for k in ("cause_label", "highlight_label",
"reasons")},
                   "software": {k: i["software"].get(k) for k in ("label", "strength_label",
"reasons")}}
                  for i in case["items"]]

response = client.messages.create(
    model="claude-opus-5-5",
    max_tokens=16000,
    system="You write claim-ready summaries of WELA results. Use only the facts given, cite power-cycle
IDs, "
           "and never change or second-guess a verdict.",
    messages=[{"role": "user", "content": json.dumps(brief)}],
)
summary = next(b.text for b in response.content if b.type == "text")
```

Phased plan

Phase	Scope	Success measure
1	Claim summary writer, pilot on closed claims, engineer approves every summary	Summaries rated correct; time saved per claim
2	"Ask the case" chat in the UI, tool access to WELA's local API	Engineers find evidence faster; answers always cite events
3	Triage at scale + confidence calibration from closed claims	Triage agrees with engineers; calibrated confidence
4	Fleet patterns across claims	New, confirmed drive / firmware / platform patterns

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A Appendix: event reference

> Every event WELA reads, as configured in `catalog.py` on this branch.

Power signals

Provider · ID	Meaning
Kernel-General 12 · EventLog 6005 / 6009	OS started · event log started · OS version at boot
Kernel-Boot 27	Boot type: 0 cold boot · 1 hybrid boot (Fast Startup) · 2 resume from hibernate
User32 1074	Shutdown / restart requested — process, reason, type, comment, user
Kernel-General 13 · EventLog 6006	OS stopped · event log stopped
Kernel-Power 109 · Winlogon 7002	Kernel shutdown initiated · user logoff
Kernel-Power 41 · EventLog 6008	Unclean shutdown reported (BugcheckCode, PowerButtonTimestamp, SleepInProgress) · previous shutdown unexpected
WER-SystemErrorReporting / BugCheck 1001	Bugcheck (crash) with its code

Provider · ID	Meaning
Kernel-Power 42 · 107 · Power-Troubleshooter 1	Entering sleep (TargetState) · resumed · sleep/wake report with times and wake source
Kernel-Power 506 · 507	Modern Standby enter · exit
Kernel-General 1 · EventLog 6013 · Eventlog 104 / 1102	System time changed · uptime · log cleared

Storage events

Provider · ID	Meaning	State
disk 7 · 11 · 51 · 52 · 154	Bad block · controller error · paging I/O error · predicted failure (SMART) · I/O failed (hardware)	error
disk 153	I/O retried	slow
disk 157	Disk surprise-removed	missing
stornvme 11	NVMe controller error	error
stornvme · storahci · iaStorAC · iaStorA · iaStorVD 129	Reset to device (incl. Intel RST / VMD)	reset
Ntfs 50 · 55 · 98 · 140	Delayed write failed · file-system corruption · volume needs repair · failed to flush log	error
volmgr 161 · 46	Dump file creation failed · crash-dump init failed	error · warning
WHEA-Logger 17 · 19 · 47	Corrected PCIe · hardware · memory error (platform)	warning
WHEA-Logger 18 · 1	Fatal hardware error · WHEA hardware error (platform)	error
StorPort (every ID)	All Storport Operational and Health events; state from level (critical/error → error, warning → warning, else info)	by level
Partition 1006	Disk inventory: DiskNumber, Model, SerialNumber, Revision, BusType, Capacity, ParentId	info
Other storage providers	Any provider whose name contains stornvme, storahci, storport, nvme, iastor or disk, at warning or above	by level

Software events

Provider · ID	Meaning
Application Error 1000 · .NET Runtime 1026	Application crashed (app, module, exception code) · .NET application crashed
Application Hang 1002	Application hung
Windows Error Reporting 1001	Error report — refined by event name: BlueScreen, LiveKernelEvent, APPCRASH / BEX, AppHang …
WindowsUpdateClient 19 · 20 · 21 · 22 · 43	Update installed · failed · needs a restart · restart scheduled · installation started
Servicing 2 · 4	Windows package installed · needs a restart
MsiInstaller 1033–1037 · 11707 · 11724	Software installed / removed / reconfigured / update installed or removed · completed
MsiInstaller 11708 · 1038 · 1005 · 1040 / 1042	Installation failed · needs a restart · restarted the PC · transaction started / ended
Service Control Manager 7031 · 7034 · 7023 · 7024	Service crashed (7031: recovery action, incl. "restart the computer") · stopped with an error

Provider · ID	Meaning
Service Control Manager 7043 · 7022 · 7011	Service held up shutdown · hung on start · timed out
Display 4101 · Resource-Exhaustion-Detector 2004 · Kernel-Power 105	Display driver stopped responding · low on memory · power source changed (AC / battery)

B Appendix: glossary

Term	Meaning
Power cycle	One transition from "on" to "off" and back: a shutdown, restart or hard shutdown closing a boot session, or a sleep / hibernate episode.
Boot session	Everything Windows logged between two boots.
Last sign of life	The last event any log recorded before the next boot — the anchor for "before power-off" windows.
Hard shutdown	The session ended without a clean stop: crash, power loss, power-button long-press, or a hang.
Fast Startup	Windows "shutdown" that hibernates the kernel; the next boot is a hybrid boot (Kernel-Boot 27, BootType 1).
Modern Standby	Low-power always-connected sleep (S0ix), logged as Kernel-Power 506 / 507.
Bugcheck	A Windows crash ("blue screen"), identified by a code such as 0x7A.
Kernel-Power 41 / 6008	Events written at the next boot when the previous shutdown was not clean.
User32 1074	The event that records who asked Windows to shut down or restart, and why.
Reason code	The numeric reason in 1074 (Windows `SHTDN_REASON_*`): planned flag, major and minor reason; independent of language.
Storport	Windows' storage port driver framework; its Operational log records timeouts, resets and latency for each drive, and WELA reads its Health log the same way.
stornvme / storahci	Windows' NVMe and SATA (AHCI) drivers. Event 129 = "reset to device".
WHEA	Windows Hardware Error Architecture — corrected and fatal hardware errors (PCIe, CPU, memory).
Partition / Diagnostic 1006	Disk inventory event logged at every boot: disk number, model, serial, firmware, bus type.
EUI-64	A 64-bit NVMe identifier that some Windows builds report in place of the serial number.
Drive window	The 60 s before power-off and the 60 s after boot that WELA checks for storage events.
App / update list	The app, update and service events shown with a power cycle: from the boot / wake that ended the one before to its power-off ("before"), and from its boot / wake to the next power-off ("after").
Needs review	Confidence below 60 % or any warning flag on a verdict.
SMART Unsafe Shutdowns	The drive's own counter of power losses without a clean shutdown notification.
LLM	Large language model — the kind of AI proposed in chapter 19 for summaries and questions.

C Appendix: commands and files

Run

```
git checkout samsung-drive-and-app-triggers
python run_wela.py                                  # UI at http://127.0.0.1:8765 (next free port if
busy)
python -m wela serve --port 8765 --no-browser
python -m wela analyze <folder or files> --out report.html --tz 330
python -m wela demo demo_logs
python -m unittest discover -s tests -v             # 51 tests
```

Collect logs on the customer PC (administrator command prompt)

```
wevtutil epl System System.evtx
wevtutil epl Application Application.evtx
wevtutil epl Microsoft-Windows-Storage-Storport/Operational Storport.evtx
wevtutil epl Microsoft-Windows-Storage-Storport/Health Storport-Health.evtx
wevtutil epl Microsoft-Windows-Partition/Diagnostic Partition.evtx   # optional: add as a file
wevtutil epl Setup Setup.evtx                                         # optional: add as a file
```

Repository layout

```
run_wela.py            start the UI
wela/
  parser.py            .evtx (wevtutil / PowerShell / python-evtx) and .xml -> events;
                       which logs a folder gives
  catalog.py           event IDs, weights, thresholds, drive vendors, software events  <- tune here
  engine.py            timeline, integrity, boots, sessions, classification, sleep,
                       drive identity + correlation, software trigger
  report.py            HTML / Excel / CSV / JSON export (no dependencies)
  server.py            local HTTP server + API
  store.py             cases saved under ./cases
  demo.py              synthetic demo logs with expected verdicts
web/                   UI (vanilla HTML/CSS/JS, works offline)
tests/                 unittest suite (demo scenario, rule tests, robustness test)
docs/                  team walkthrough (.pptx / .pdf), this reference and its source (src/)
```

This release (branch samsung-drive-and-app-triggers)

Update of 1 November 2026

- **App / update lists run cycle to cycle:** from the power cycle before to this one and on to the next — 100 nearest first, the rest on request; scroll boxes open at the end nearest the power cycle; "Earlier, since …" in the reasons.
- **Search by event ID** in Power cycles, Drive, Apps & updates and the detail drawer; **Near** shows the power cycles before and after, clickable.
- **Folders:** only System, Application and Storport (Operational, Health) are read; skipped logs are counted.
- **Simpler:** the SMART check tab is gone; the home screen no longer fills in the last path.
- **Quality:** 4 new tests (51 total).

October 2026

Which drive — every storage event tied to a drive, Samsung highlighted across UI, timeline, HTML and Excel; **what triggered it** — 1074 decoding, Windows Update, installers, OEM / IT tools, crashes and hangs, with confidence and the symptom-vs-cause check; the Apps & updates tab and the Excel software sheets; 27 new tests (47 total) and a demo case of 16 power cycles.